BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 156 · Lesson 156 · Client-Side versus Server-Side Processing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does PageRank divide a page's score among outgoing links?

Q2. What data structure follows LIFO?

Q3. What is a database foreign key?

Q4. What is the role of a router?

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I Do · Read new knowledge

Key vocabulary

client-side processing — Computing performed in browser/device.

server-side processing — Computing performed by backend.

authentication — Checking claimed identity.

authorisation — Checking what a user may do.

New learning

Browser tasks — JavaScript can update local interface promptly.

Backend tasks — Server authenticates, authorises and validates record changes.

Compare deployment — Balance speed, privacy, load and security boundaries.

Worked example

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system protected?

1. Browser value is untrusted — Identify the input or diagram and predict the first result.

2. Server checks permissions — Follow the second stage and explain its importance.

3. Reject forbidden edit — The server must authenticate the user and enforce edit permissions, ignoring unauthorised mark values. It validates requests and writes to the authoritative database only after checking access.

We Do · Practise with guidance

A public revision quiz shows hints instantly but stores no marks. Which processing placement fits?

Public quiz

Identify the input or diagram and predict the first result.

Browser computes feedback

Follow the second stage and explain its importance.

No private data stored

Client-side JavaScript can show hints immediately with fewer server requests; if restricted answers or persistent records matter, keep them server-controlled.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give two benefits and one limitation of client-side processing. [6]

Question 2

Explain why permission to change assessment results must be checked server-side. [6]

Question 3

A hub offers instant quiz hints, private reports and offline revision. Assign client/server roles and justify. [8]

Exit Ticket · Answer before checking

Browser validation is sufficient to authorise database changes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Each outgoing link gets a share of source importance.

Q2. Stack.

Q3. A reference to a key in another table.

Q4. Forward packets between IP networks.

You Do pages · Independent questions

Q1. Fast feedback [2], reduced server requests [2], tampering/limited client resources [2].

Q2. Client untrusted [2], authorisation check [2], database change controlled centrally [2].

Q3. Public local feedback [2], authenticated server reports [2], cached offline resources [2], balanced security/latency [2].

Exit Ticket · Model answer

False. Server must independently authorise.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: